

OVERVIEW

The Power Board turns a USB-C input and a single Li-Po cell into the clean, always-available 5 V that the Main Board runs on. It charges and manages the battery through a power-path front end, boosts the cell up to a regulated 5 V, and distributes it to the Main Board over an FPC, while also feeding the device's surrounding LED arc and monitoring battery voltage.

AT A GLANCE

Input USB-C, 5 V	Battery 3.7 V rechargeable Li-Po cell
Charger BQ24075 · Li-Ion/Li-Po, power-path (DPPM)	Boost TPS61022 · high-current synchronous 5 V boost
Outputs 5V_VDD to Main Board (FPC) · 5V_LED (arc)	Monitoring battery-voltage analog sense (ADC)

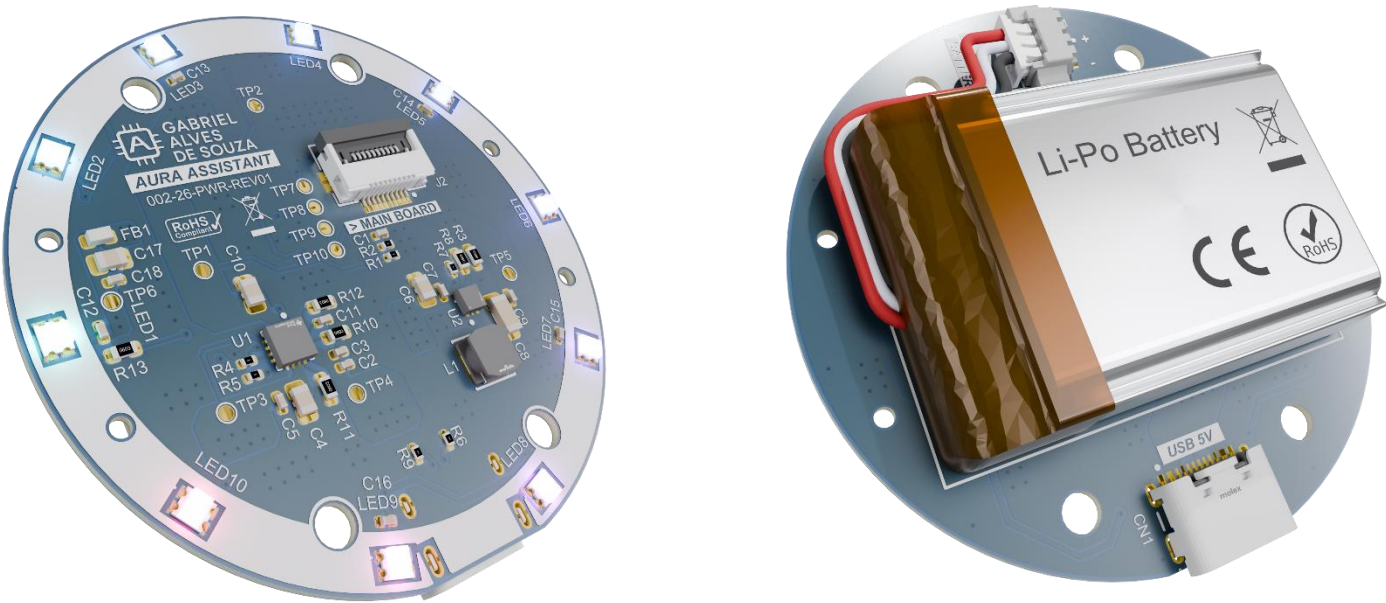
SYSTEM BLOCKS

- **USB-C input** — input conditioning & protection
- **Li-Po charger** — power-path front end (BQ24075)
- **5 V boost** — high-current synchronous converter (TPS61022)
- **Distribution** — rail routing & FPC link to Main Board
- **LED-arc supply** — 5V_LED for the surrounding arc
- **Battery telemetry** — voltage sense for state-of-charge

ENGINEERING HIGHLIGHTS

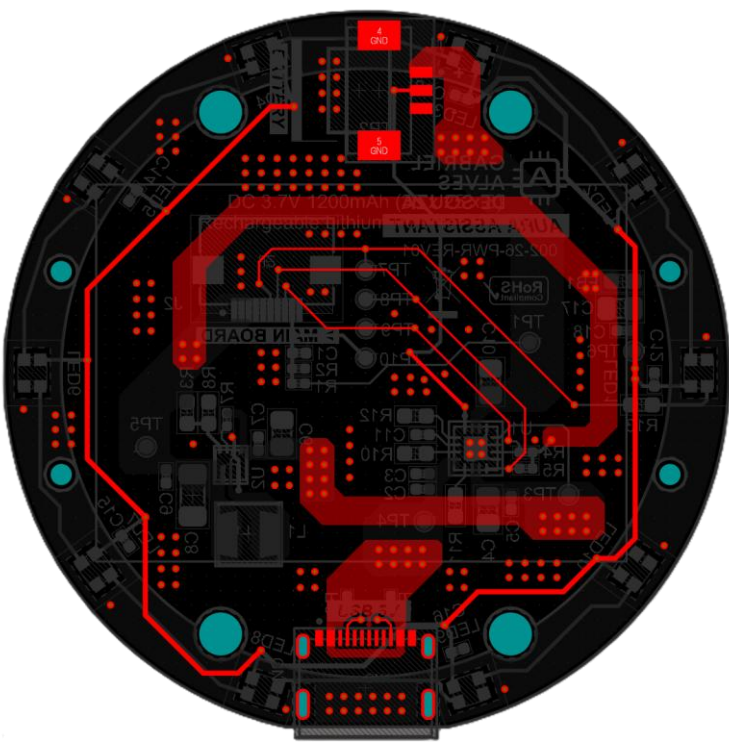
- **Power-path charging:** the device keeps running from USB while the battery charges, with a seamless hand-off to battery when unplugged.
- **Battery-to-5 V boost** sized for the full system load, delivering a stable rail to the Main Board across the FPC.
- **High-current layout:** short, wide copper on the charge and boost paths with tight switching loops for efficiency and low EMI.
- **Battery-voltage sensing** for state-of-charge and safe operation.

PCB 3D View

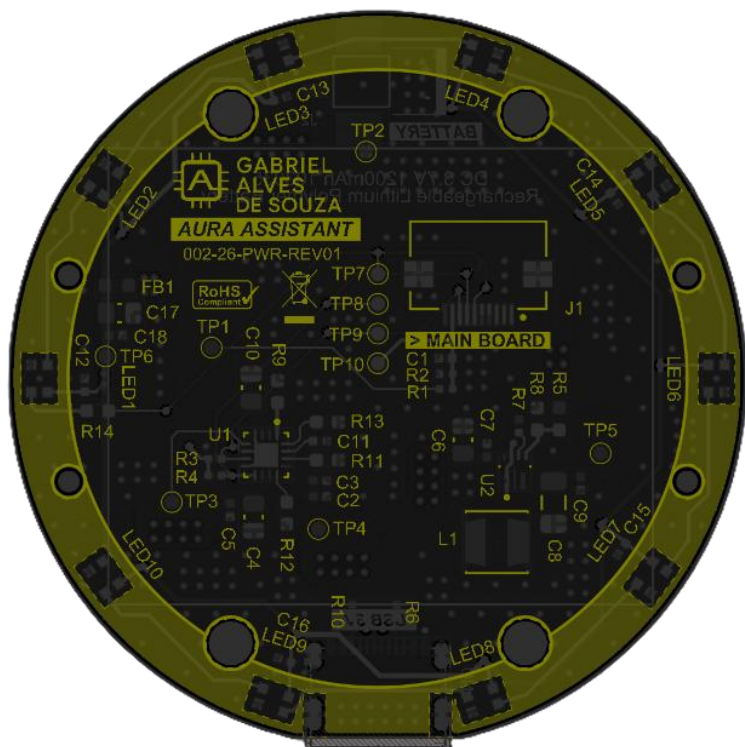
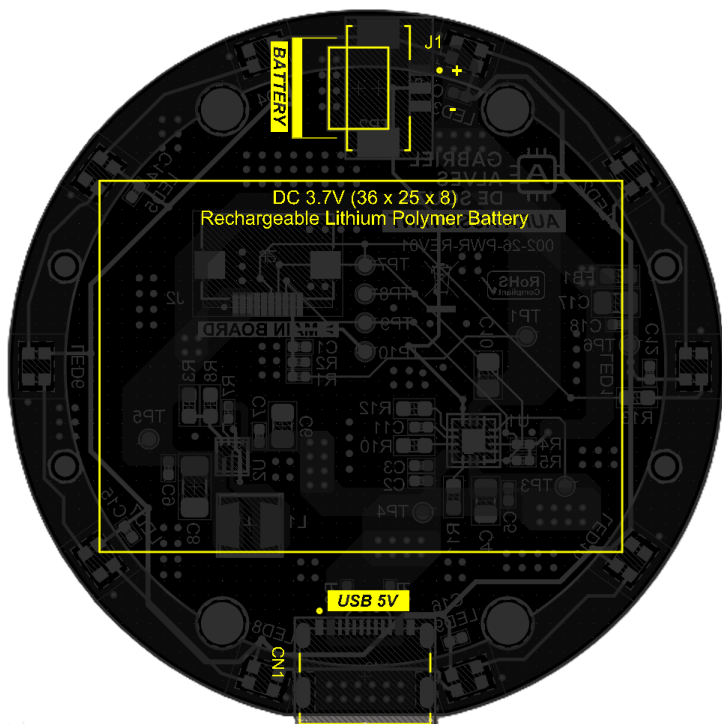
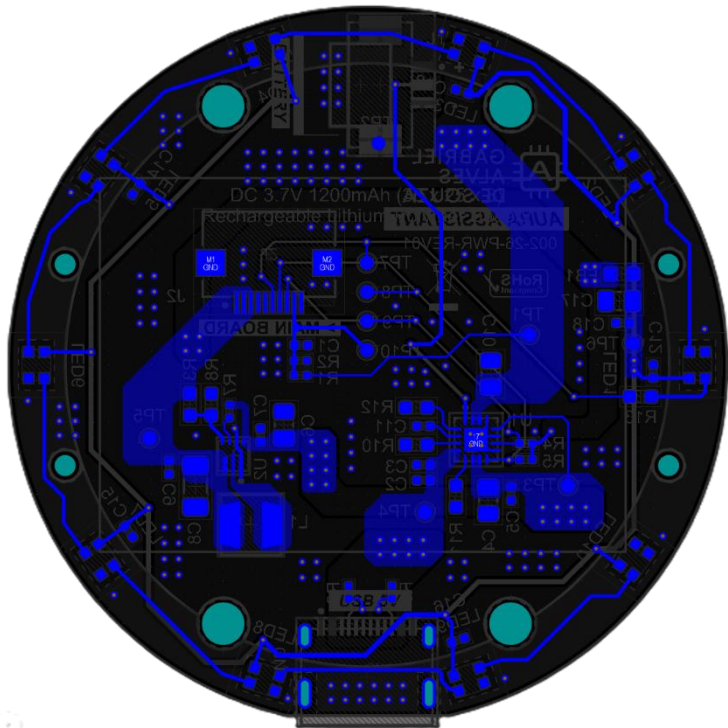


Altium Designer PCB Screenshots

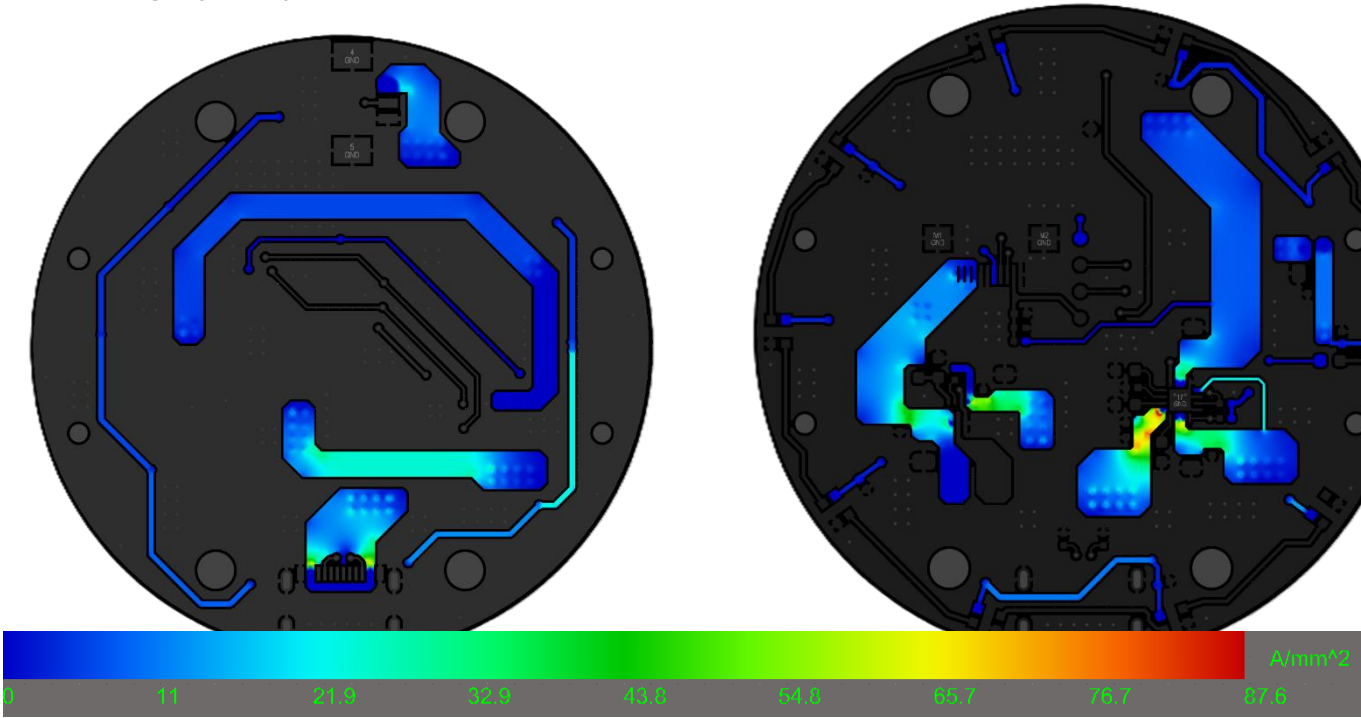
L1 – Top Layer



L2 – Bottom



Power Integrity Analysis



DC Drop Limit: 250mV

Passed

Voltage Drop: 11.475mV	Current Density: 87.6A/mm ²	Max Via Current: 428.908mA
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PCB 3D View

